

[Dashboard](#) / [My courses](#) / [VL MLPC \(2022S\)](#) / [Final Exam](#) / [Exam](#)**Started on** Monday, 13 June 2022, 12:12 PM**State** Finished**Completed on** Monday, 13 June 2022, 1:37 PM**Time taken** 1 hour 25 mins**Marks** 71.00/79.00**Grade** 89.87 out of 100.00

Information

1. Bayesian Classification

Consider the following decision strategies for classifying an object (feature vector) $\mathbf{x}$ in a two-class problem with classes ω_1, ω_2 :

- S1. Predict ω_1 iff $P(\omega_1) > P(\omega_2)$
- S2. Predict ω_1 iff $p(\mathbf{x} | \omega_1) > p(\mathbf{x} | \omega_2)$
- S3. Predict ω_1 iff $P(\omega_1 | \mathbf{x}) > P(\omega_2 | \mathbf{x})$
- S4. Predict ω_1 iff $P(\omega_1) > 0.5$
- S5. Predict ω_1 iff $p(\mathbf{x} | \omega_1) > p(\mathbf{x})$
- S6. Predict ω_1 iff $P(\omega_1 | \mathbf{x}) > P(\omega_1)$

and assume that we know the true values of the various probabilities appearing above.

Question 1

Correct

Mark 2.00 out of 2.00

Which of these strategies result in the baseline accuracy?

- ☒ S1 ✓
- ☐ S2
- ☐ S3
- ☒ S4 ✓
- ☐ S5
- ☐ S6

Die Antwort ist richtig.

The correct answers are:

S1,

S4

Question **2**

Correct

Mark 2.00 out of 2.00

Which of these strategies are guaranteed to have the Bayes Error?

- ☐ S1
- ☐ S2
- ☒ S3 ✓
- ☐ S4
- ☐ S5
- ☐ S6

Die Antwort ist richtig.

The correct answer is:

S3

Question **3**

Partially correct

Mark 1.00 out of 2.00

Which of these strategies take into account both prior and likelihood?

- ☐ S1
- ☐ S2
- ☒ S3 ✓
- ☐ S4
- ☐ S5
- ☐ S6

Die Antwort ist teilweise richtig.

You have correctly selected 1.

The correct answers are:

S3,

S6

Information

Under which conditions will S2 and S3 make identical predictions?

Question **4**

Correct

Mark 1.00 out of 1.00

$$P(\omega_1) = 0$$

Select one:

- ☐ True
- ☒ False ✓

The correct answer is 'False'.

Question **5**

Correct

Mark 1.00 out of 1.00

$$P(\omega_1) = P(\omega_2)$$

Select one:

- ☒ True ✓
- ☐ False

The correct answer is 'True'.

Question **6**

Correct

Mark 1.00 out of 1.00

$$p(\mathbf{x} \mid \omega_1) = p(\mathbf{x} \mid \omega_2)$$

Select one:

- ☐ True
- ☒ False ✓

The correct answer is 'False'.

Question **7**

Correct

Mark 1.00 out of 1.00

$$P(\omega_1) = 1$$

Select one:

- ☐ True
- ☒ False ✓

The correct answer is 'False'.

Information

Which of the following statements are true?

Question 8

Correct

Mark 1.00 out of 1.00

If the two classes are equally frequent ($P(\omega_1) = P(\omega_2)$), then S2 and S3 are equivalent.

Select one:

- ☒ True ✓
- ☐ False

The correct answer is 'True'.

Question 9

Correct

Mark 1.00 out of 1.00

If $P(\omega_1 \mid \mathbf{x}) > P(\omega_1)$, then $P(\omega_2 \mid \mathbf{x}) < P(\omega_2)$.

Select one:

- ☒ True ✓
- ☐ False

The correct answer is 'True'.

Question 10

Incorrect

Mark 0.00 out of 1.00

If $p(\mathbf{x} \mid \omega_1) > p(\mathbf{x})$, then $p(\mathbf{x} \mid \omega_2) < p(\mathbf{x})$.

Select one:

- ☐ True
- ☒ False ✗

The correct answer is 'True'.

[◀ Lecture 11](#)[Exam Procedure and Rules ▶](#)

[Dashboard](#) / [My courses](#) / [VL MLPC \(2022S\)](#) / [Final Exam](#) / [Exam](#)

Information

2. Classifier Evaluation (a)

Consider an arbitrary classifier X for a binary classification problem, with unknown classification behavior. If this classifier were evaluated on a test set T consisting **only of positive examples** (i.e., if the test examples all belong to the positive class), which of the following statements would be correct, and which would be wrong:

The accuracy of X , as measured on T , can be expected to be ...

Question 11

Correct

Mark 1.00 out of 1.00

... equal to the baseline accuracy (on T).

Select one:

- ☐ True
- ☒ False ✓

The correct answer is 'False'.

Question 12

Correct

Mark 1.00 out of 1.00

... 0.0

Select one:

- ☐ True
- ☒ False ✓

The correct answer is 'False'.

Question 13

Correct

Mark 1.00 out of 1.00

... equal to its recall (on T).

Select one:

- ☒ True ✓
- ☐ False

The correct answer is 'True'.

Question **14**

Correct

Mark 1.00 out of 1.00

... equal to the accuracy (on T) of a classifier that always predicts the positive class.

Select one:

- ☐ True
- ☒ False ✓

The correct answer is 'False'.

Question **15**

Correct

Mark 1.00 out of 1.00

... 1.0 (100%)

Select one:

- ☐ True
- ☒ False ✓

The correct answer is 'False'.

Information

2. Classifier Evaluation (b)

Consider a 2-class classification problem with classes p and n , and a 2D "**Recall-Precision (RP) Space**" (in analogy to ROC space), where the horizontal axis (x) is recall on class p , and the vertical axis (y) is precision on class p .

Which points in this RP space correspond to the following default classifiers:

Question **16**

Correct

Mark 1.00 out of 1.00

All p classifier (a classifier that always predicts class p):

- ☐ lower left corner
- ☐ upper left corner
- ☒ upper right corner ✓
- ☐ lower right corner
- ☐ possibly some other point in the space
- ☐ one of the measures is undefined

Die Antwort ist richtig.

The correct answers are:

upper right corner,

possibly some other point in the space

Question **17**

Incorrect

Mark 0.00 out of 1.00

All n classifier:

- ☒ lower left corner ✗
- ☐ upper left corner
- ☐ upper right corner
- ☐ lower right corner
- ☐ possibly some other point in the space
- ☐ one of the measures is undefined

Die Antwort ist falsch.

The correct answer is:

one of the measures is undefined

Question 18

Correct

Mark 1.00 out of 1.00

Random 50:50 guessing classifier:

- ☐ point ($x = 1.0; y = 0.5$)
- ☐ point ($x = 0.5; y = 1.0$)
- ☐ point ($x = 0.5; y = 0.5$)
- ☒ possibly some other point in the space ✓
- ☐ one of the measures is undefined

Die Antwort ist richtig.

The correct answers are: point ($x = 0.5; y = 0.5$), possibly some other point in the space

Information

2. Classifier Evaluation (c)

Which of the following statements are true, and which are false (always relative to some fixed dataset D)?

Question 19

Correct

Mark 1.00 out of 1.00

A classifier that predicts between the two classes randomly will have Recall = Precision = 0.5.

Select one:

- ☐ True
- ☒ False ✓

The correct answer is 'False'.

Question 20

Incorrect

Mark 0.00 out of 1.00

Two classifiers with the same recall and same precision can have different accuracy.

Select one:

- ☒ True ✗
- ☐ False

The correct answer is 'False'.

Question **21**

Correct

Mark 1.00 out of 1.00

If a classifier A has higher recall than B on a dataset D, it must have lower precision than B.

Select one:

- ☐ True
- ☒ False ✓

The correct answer is 'False'.

◀ Lecture 11

Jump to...

Exam Procedure and Rules ▶

[Dashboard](#) / [My courses](#) / [VL MLPC \(2022S\)](#) / [Final Exam](#) / [Exam](#)

Information

3. Classifiers

Assume a binary classification task with classes $\Omega = \{\text{salmon}, \text{bass}\}$ and two (heavily discretized) features x_1 (*Length*) and x_2 (*Lightness*).

Also assume we have a training set $\mathcal{D}$ consisting of four fish only:

$[\text{short}, \text{dark}], \text{salmon}$
 $[\text{short}, \text{light}], \text{salmon}$
 $[\text{long}, \text{dark}], \text{bass}$
 $[\text{long}, \text{light}], \text{bass}$

3. Classifiers (a)

Question **22**

Correct

Mark 2.00 out of 2.00

Calculate the empirical entropy (log base 2) of the class distribution in $\mathcal{D}$ (no calculator needed).

Answer: ✓

The correct answer is: 1

Question **23**

Correct

Mark 2.00 out of 2.00

Calculate the information gains of the two features *Length* and *Lightness* on $\mathcal{D}$ (no calculator needed).

Answer: ✓

The correct answer is: 1



Question 24

Correct

Mark 1.00 out of 1.00

What is the depth of the simplest decision tree consistent with this dataset? (The depth of a tree that only consists of 1 node (the root) is 0; the depth of an arbitrary tree is the length of its longest branch).

Answer: ✓

The correct answer is: 1

Question 25

Correct

Mark 1.00 out of 1.00

On this dataset, would the ID3 algorithm be guaranteed to find the simplest possible tree?

Select one:

- ☒ True ✓
☐ False

The correct answer is 'True'.

Question 26

Correct

Mark 3.00 out of 3.00

Specify the probability distributions that Naive Bayes would use for classification, given this dataset for training.

$P(\text{Class}) = [\text{salmon} = 0.5 \text{ } \checkmark, \text{bass} = 0.5 \text{ } \checkmark]$

$P(\text{Length} \mid \text{Class} = \text{salmon}) = [\text{short} = 1 \text{ } \checkmark, \text{long} = 0 \text{ } \checkmark]$

$P(\text{Length} \mid \text{Class} = \text{bass}) = [\text{short} = 0 \text{ } \checkmark, \text{long} = 1 \text{ } \checkmark]$

$P(\text{Lightness} \mid \text{Class} = \text{salmon}) = [\text{dark} = 0.5 \text{ } \checkmark, \text{light} = 0.5 \text{ } \checkmark]$

$P(\text{Lightness} \mid \text{Class} = \text{bass}) = [\text{dark} = 0.5 \text{ } \checkmark, \text{light} = 0.5 \text{ } \checkmark]$

Information

If we added 5 copies of the fourth example ([long, light]; bass) as additional training examples to $\mathcal{D}$, which of the following statements would be true?

Question **27**

Correct

Mark 1.00 out of 1.00

The decision tree learned by ID3 would degenerate to a root node that predicts the majority class (bass).

Select one:

- ☐ True
- ☒ False ✓

The correct answer is 'False'.

Question **28**

Correct

Mark 1.00 out of 1.00

The bass examples from D would end up in the same leaf.

Select one:

- ☒ True ✓
- ☐ False

The correct answer is 'True'.

Question **29**

Correct

Mark 1.00 out of 1.00

The prior for Naive Bayes would change.

Select one:

- ☒ True ✓
- ☐ False

The correct answer is 'True'.

Question **30**

Correct

Mark 1.00 out of 1.00

The decision tree learned by ID3 would get one level deeper.

Select one:

- ☐ True
- ☒ False ✓

The correct answer is 'False'.



Question 31

Correct

Mark 1.00 out of 1.00

The decision tree learned by ID3 would stay exactly the same.

Select one:

- ☒ True ✓
- ☐ False

The correct answer is 'True'.

Information

3. Classifiers (b)

Now suppose we encode the features and class labels in the above dataset as numbers (short = 0, long = 1; dark = 0, light = 1; salmon = 0, bass = 1).

Question 32

Correct

Mark 1.00 out of 1.00

Are the classes linearly separable?

Select one:

- ☒ True ✓
- ☐ False

The correct answer is 'True'.

Question 33

Correct

Mark 1.00 out of 1.00

How many units do we need to express a classifier (without hidden layer) that predicts $P(\text{bass} \mid \text{Features})$ as a feed-forward neural network? (Don't forget the bias unit!)

Answer: ✓

The correct answer is: 4



Question 34

Incorrect

Mark 0.00 out of 1.00

Which examples in $\mathcal{D}$ are exactly on the margin (i.e., are support vectors) in a maximum margin linear classifier?

- ☐ ([short, dark], salmon), ([short, light], salmon)
- ☐ ([short, dark], salmon), ([long, light], bass)
- ☐ ([short, dark], salmon)
- ☐ all four
- ☒ none of them ✖
- ☐ a maximum margin linear classifier does not exist for this dataset

Die Antwort ist falsch.

The correct answer is:
all four

Information

If we added four (4) copies of the fourth example ([long, light]; bass) as additional training examples to $\mathcal{D}$, which of the following statements would be true?

Question 35

Correct

Mark 1.00 out of 1.00

The decision boundary for a k -NN classifier with $k = 4$ would look different than the boundary with $k = 1$.

Select one:

- ☒ True ✔
- ☐ False

The correct answer is 'True'.

Question 36

Correct

Mark 1.00 out of 1.00

A feed-forward neural network for this dataset would require twice as many input units (not considering the bias unit).

Select one:

- ☐ True
- ☒ False ✔

The correct answer is 'False'.



Question **37**

Correct

Mark 1.00 out of 1.00

A k-NN classifier with $k = 1$ would change its prediction behaviour.

Select one:

- ☐ True
- ☒ False ✓

The correct answer is 'False'.

Question **38**

Correct

Mark 1.00 out of 1.00

A k-NN classifier with $k = 9$ would always predict class bass.

Select one:

- ☒ True ✓
- ☐ False

The correct answer is 'True'.

[◀ Lecture 11](#)[Exam Procedure and Rules ▶](#)

[Dashboard](#) / [My courses](#) / [VL MLPC \(2022S\)](#) / [Final Exam](#) / [Exam](#)

Information

4. Neural Networks and Deep Learning (a)

Consider the following four possible loss functions for a regression problem, where

- y_i ... true target value for example $\mathbf{x}_i$
- $\hat{y}(\mathbf{x}_i)$... value predicted by model from input vector $\mathbf{x}_i$

Which of these measures are **not** meaningful as error measures on a numeric regression task?

Question **39**

Correct

Mark 1.00 out of 1.00

$$L_1 = \frac{1}{|\mathcal{D}|} \sum_{\mathbf{x}_i \in \mathcal{D}} (y_i - \hat{y}(\mathbf{x}_i))$$

Not meaningful?

Select one:

- ☒ True ✓
- ☐ False

The correct answer is 'True'.

Question **40**

Correct

Mark 1.00 out of 1.00

$$L_4 = \frac{1}{|\mathcal{D}|} \sum_{\mathbf{x}_i \in \mathcal{D}} e^{|y_i - \hat{y}(\mathbf{x}_i)|}$$

Not meaningful?

Select one:

- ☐ True
- ☒ False ✓

The correct answer is 'False'.

Question **41**

Correct

Mark 1.00 out of 1.00

$$L_2 = \frac{1}{|\mathcal{D}|} \sum_{\mathbf{x}_i \in \mathcal{D}} |y_i - \hat{y}(\mathbf{x}_i)|$$

Not meaningful?

Select one:

- ☐ True
- ☒ False ✓

The correct answer is 'False'.

Question **42**

Correct

Mark 1.00 out of 1.00

$$L_3 = \frac{1}{|\mathcal{D}|} \sum_{\mathbf{x}_i \in \mathcal{D}} (y_i - \hat{y}(\mathbf{x}_i))^2$$

Not meaningful?

Select one:

- ☐ True
- ☒ False ✓

The correct answer is 'False'.

Information

4. Neural Networks and Deep Learning (b)

Weight Learning by Gradient Descent: Which of the following statements are true?

Question **43**

Correct

Mark 1.00 out of 1.00

A loss function can also be defined over several (more than one) output units.

Select one:

- ☒ True ✓
- ☐ False

The correct answer is 'True'.

Question **44**

Correct

Mark 1.00 out of 1.00

A gradient of zero will result in unchanged weights.

Select one:

- ☒ True ✓
- ☐ False

The correct answer is 'True'.

Question **45**

Incorrect

Mark 0.00 out of 1.00

A large learning rate may lead to convergence more quickly.

Select one:

- ☐ True
- ☒ False ✗

The correct answer is 'True'.

Question **46**

Incorrect

Mark 0.00 out of 1.00

The gradient must be calculated for each weight in the network.

Select one:

- ☐ True
- ☒ False ✗

The correct answer is 'True'.

Question **47**

Correct

Mark 1.00 out of 1.00

A loss function can have more than one minimum.

Select one:

- ☒ True ✓
- ☐ False

The correct answer is 'True'.

Question **48**

Correct

Mark 1.00 out of 1.00

A small learning rate leads to more complex neural network models.

Select one:

- ☐ True
- ☒ False ✓

The correct answer is 'False'.

Question **49**

Correct

Mark 1.00 out of 1.00

The gradient is a number in the range $[-1,1]$.

Select one:

- ☐ True
- ☒ False ✓

The correct answer is 'False'.

Information

4. Neural Networks and Deep Learning (c)

Overfitting Avoidance: Which of the following strategies could help prevent overfitting in a neural network?

Question **50**

Correct

Mark 1.00 out of 1.00

Decrease the number of hidden units.

Select one:

- ☒ True ✓
- ☐ False

The correct answer is 'True'.

Question **51**

Correct

Mark 1.00 out of 1.00

Decrease the number of hidden layers.

Select one:

- ☒ True ✓
- ☐ False

The correct answer is 'True'.

Question **52**

Correct

Mark 1.00 out of 1.00

Reduce learning rate.

Select one:

- ☐ True
- ☒ False ✓

The correct answer is 'False'.

Question **53**

Correct

Mark 1.00 out of 1.00

Reduce the number of training iterations.

Select one:

- ☒ True ✓
- ☐ False

The correct answer is 'True'.

Question **54**

Correct

Mark 1.00 out of 1.00

Limit the size of gradient; crop if too large.

Select one:

- ☐ True
- ☒ False ✓

The correct answer is 'False'.

Question **55**

Correct

Mark 1.00 out of 1.00

Change weights in direction of the gradient (instead of in the negative direction).

Select one:

- ☐ True
- ☒ False ✓

The correct answer is 'False'.

[◀ Lecture 11](#)

Jump to...

[Exam Procedure and Rules ▶](#)

[Dashboard](#) / [My courses](#) / [VL MLPC \(2022S\)](#) / [Final Exam](#) / [Exam](#)

Question **56**

Correct

Mark 2.00 out of 2.00

5. Unsupervised Learning

Consider the k-means clustering algorithm as described in the lecture slides.

5. Unsupervised Learning (a)

What qualities of a cluster partitioning does the Mean Quantisation Error (MQE) implicitly try to encode?

- ☐ The average size of the cluster centers
- ☒ The compactness of the clusters ✓
- ☐ The average distance between cluster centers
- ☐ Equal distribution of examples among clusters
- ☐ The number of resulting clusters

Your answer is correct.

The correct answer is:

The compactness of the clusters

Question 57

Correct

Mark 2.00 out of 2.00

5. Unsupervised Learning (b)

Which of the following definition variants of the Mean Quantisation Error (MQE) are equivalent, in the sense of leading to exact same behavior of the k-means algorithm?

$$\frac{1}{|\mathcal{D}|} \sum_{\mathbf{x}_i \in \mathcal{D}} D(\mathbf{x}_i, c(\mathbf{x}_i))$$

$$\frac{1}{2} \sum_{\mathbf{x}_i \in \mathcal{D}} D(\mathbf{x}_i, c(\mathbf{x}_i))$$

$$\frac{1}{|\mathcal{D}|} \sum_{\mathbf{x}_i \in \mathcal{D}} e^{D(\mathbf{x}_i, c(\mathbf{x}_i))}$$

- ☒ all three are equivalent ✓
- ☐ only (2) and (3) are equivalent
- ☐ only (1) and (2) are equivalent
- ☐ none; they may all lead to different behaviour
- ☐ only (1) and (3) are equivalent

Your answer is correct.

The correct answer is:
all three are equivalent

Information

5. Unsupervised Learning (c)

Which of the following statements about the **Bayes Information Criterion (BIC)** are true (here it is once more so that you don't have to look it up):

$$BIC(M : \mathcal{D}) = \log P(\mathcal{D} | M) - p \cdot \frac{\log |\mathcal{D}|}{2}$$

Question 58

Correct

Mark 1.00 out of 1.00

If we multiplied the BIC score by some constant $c > 1$, a model selection algorithm using this score would tend to return simpler models.

Select one:

- ☐ True
- ☒ False ✓

The correct answer is 'False'.

Question **59**

Correct

Mark 1.00 out of 1.00

If we removed the second term from the formula, leaving only $\log P(\mathcal{D} \mid M)$, the resulting measure would favor simple models, without regard to how well they fit the data.

Select one:

- ☐ True
- ☒ False ✓

The correct answer is 'False'.

Question **60**

Correct

Mark 1.00 out of 1.00

The second term of the formula puts a penalty on cluster partitionings that have a large number of objects per cluster.

Select one:

- ☐ True
- ☒ False ✓

The correct answer is 'False'.

Question **61**

Correct

Mark 1.00 out of 1.00

If we removed the second term from the formula, leaving only $\log P(\mathcal{D} \mid M)$, the resulting measure would favor models that maximally fit the data.

Select one:

- ☒ True ✓
- ☐ False

The correct answer is 'True'.

[◀ Lecture 11](#)[Exam Procedure and Rules ▶](#)

[Dashboard](#) / [My courses](#) / [VL MLPC \(2022S\)](#) / [Final Exam](#) / [Exam](#)

Information

6. Hidden Markov Modles

Which of the following statements about Hidden Markov Models (HMMs) are true?

Question **62**

Correct

Mark 1.00 out of 1.00

In MAP classification of a particular observation sequence, the prior in Bayes' rule can be calculated from the HMM.

Select one:

- ☐ True
- ☒ False ✓

The correct answer is 'False'.

Question **63**

Correct

Mark 1.00 out of 1.00

The number of observation symbols must be equal to the number of states.

Select one:

- ☐ True
- ☒ False ✓

The correct answer is 'False'.

Question **64**

Correct

Mark 1.00 out of 1.00

An observation matrix **B** may contain columns that consist of zeroes only.

Select one:

- ☐ True
- ☒ False ✓

The correct answer is 'False'.

Question **65**

Correct

Mark 1.00 out of 1.00

The state transition matrix **A** is always of square shape.

Select one:

- ☒ True ✓
- ☐ False

The correct answer is 'True'.

Question **66**

Correct

Mark 1.00 out of 1.00

The Forward Algorithm calculates the probability of a state sequence.

Select one:

- ☐ True
- ☒ False ✓

The correct answer is 'False'.

Question **67**

Incorrect

Mark 0.00 out of 1.00

If, in a given HMM with N states $s_1, \dots, s_N$, if $a_{ij} = P(s_i \rightarrow s_j) = 1/N$ for all i, j , the same process could be modelled with just a single state.

Select one:

- ☒ True ✗
- ☐ False

The correct answer is 'False'.

Question **68**

Correct

Mark 1.00 out of 1.00

An observation model matrix **B** may contain zeroes.

Select one:

- ☒ True ✓
- ☐ False

The correct answer is 'True'.

Question **69**

Correct

Mark 1.00 out of 1.00

The Forward Algorithm calculates the probability of an observation sequence.

Select one:

- ☒ True ✓
- ☐ False

The correct answer is 'True'.

Question **70**

Correct

Mark 1.00 out of 1.00

A HMM permits us to calculate the probability of any state sequence (disregarding observations) of arbitrary length.

Select one:

- ☒ True ✓
- ☐ False

The correct answer is 'True'.

[◀ Lecture 11](#)[Exam Procedure and Rules ▶](#)